BEGINNER → MID-LEVEL PYTHON DEVELOPER, NO CS DEGREE

Python Career Roadmap

Yes, the program can get someone without a CS degree into the job market. It builds the skills. Three additions help those skills get past hiring filters and interviews.

It produces evidence

Junior hiring runs on proof: deployed projects, tests, Git history, and a walkthrough of your own code. Every phase ends in one of those, and the daily commits in the learning repo are evidence too.

Phase 2 is real CS

Composing Programs is the text of Berkeley's CS 61A. It covers the core ideas of a first-year computer science course: abstraction, recursion, efficiency, interpreters, and SQL.

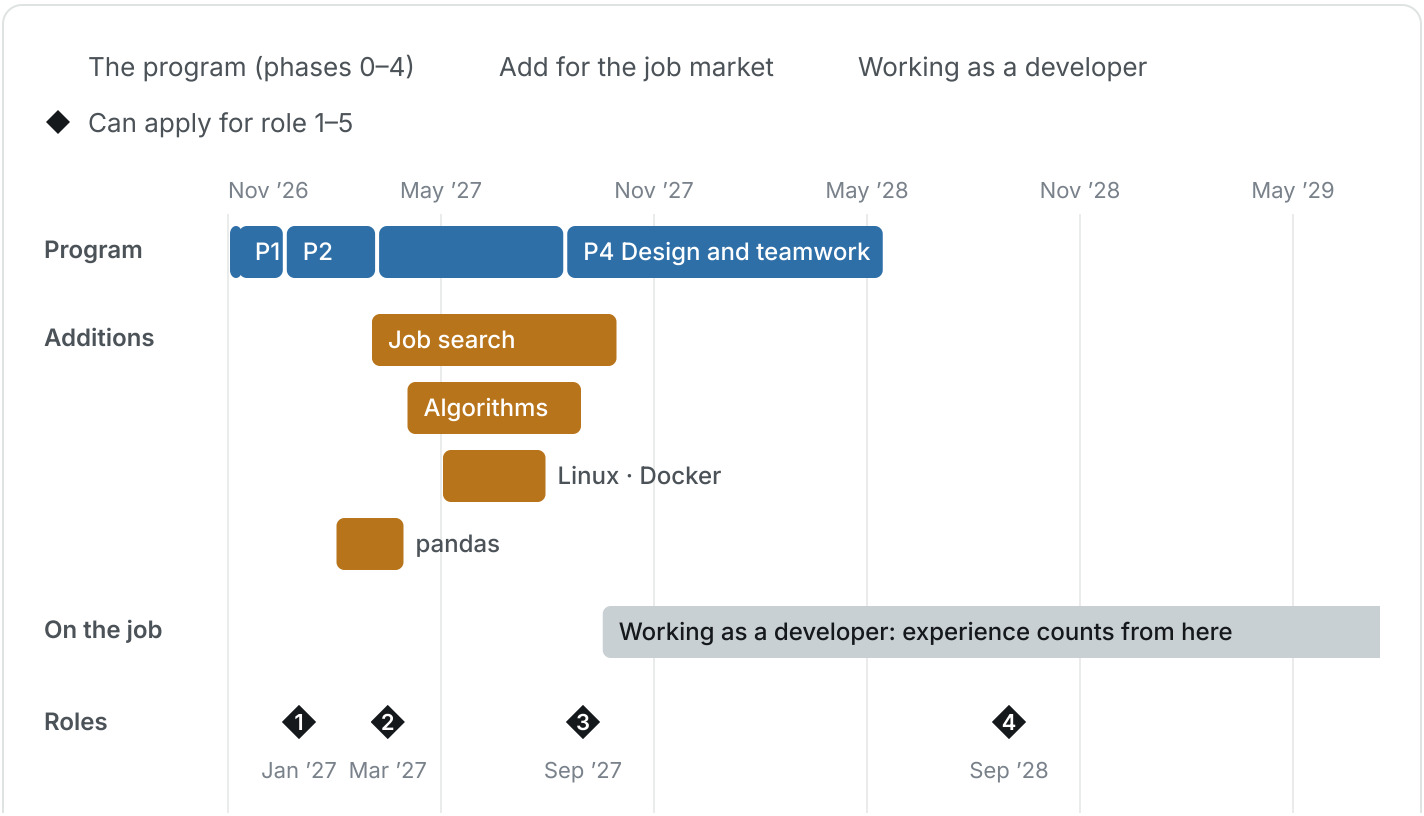
Where a degree still counts

Some large employers and most visa sponsorship screen on degrees. Referrals and internal moves skip that screen, so the student's new full-time job may be the best way in.

Titles follow experience

Studying alone doesn't make anyone mid-level. That title usually comes 2–3 years into a developer job. The plan gets the first job and makes those years count.

Milestones by month



To scale from Nov 2026. Assumes 8 hours a day until Nov 14, then 6–8 hours a week.
Hover or tab to a bar or marker for details. The same information is in the role list below.

Roles, in the order they open up

1

1. IT support with scripting · manual QA tester

Earliest Jan 2027 · after Phase 1

NEEDS

`pcc-00…pcc-11` `proj-mini` `proj-deployed-app`

SHOW

The deployed app, a GitHub history of daily commits, and a script that automates something at the current job.

INTERVIEW

Practical questions, sometimes a small take-home task.

NOTE

Easiest as an internal move at the current employer.

2

2. Junior data analyst · QA automation engineer

Earliest Apr 2027 · after Phase 2

NEEDS

`pcc-11` `pcc-15…pcc-17` `cp-4.3` `sqlbolt` `pandas`

SHOW

An analysis of real data using SQL, or a pytest suite that tests a real app.

INTERVIEW

Live SQL queries, a data or test-design task.

NOTE

Data analyst needs the pandas addition. QA automation leans on pytest.

3

3. Junior Python / backend developer

Earliest Sep 2027 · after Phase 3

NEEDS

`proj-capstone` `fp-01…fp-18` `algorithms` `linux-docker` `job-search`

SHOW

The capstone with real users, tests, CI, a database and Docker, plus two earlier projects.

INTERVIEW

Coding problems (easy to medium), a walkthrough of your project, basic HTTP and database questions.

NOTE

The first real developer title. Everything after this is driven by work experience.

4

4. Software engineer

Around Sep 2028 · about a year into the first developer job

NEEDS

`ap-01…ap-06` `proj-oss` `pragprog`

SHOW

Features shipped at work, bugs fixed in production, pull requests reviewed.

INTERVIEW

Usually a promotion. Externally: coding plus a small design discussion.

NOTE

Dropping the 'junior' label usually takes 12–18 months on the job.

5

5. Mid-level developer

Sep 2029 – late 2030 · 2–3 years into developer work

NEEDS

`PLAN.md checklist: all ticked`

SHOW

Owns features end to end, writes small design docs, mentors a junior.

INTERVIEW

System design for a small service, deep project discussion, code review exercise.

NOTE

The mid-level checklist in PLAN.md is what interviewers probe at this level.

Highlighted tags are the additions below. Plain tags are logs already in the learning repo.